Jianshu Zhang

Computer Science Ph.D. Student @ Northwestern University

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Research Summary

- I am a second-year **CS Ph.D.** student at **Northwestern University**. I study how agents can truly step into the real world, with a focus on grounding and understanding (perception, spatial capabilities, and safety boundaries), contextual reasoning, and continual learning and self-evolution, with a particular interest in **data-centric approaches** to evaluating and improving their capabilities.

Education

Ph.D. in Computer Science 2025-2030
Northwestern University Evanston, USA
Advisor: Prof. Han Liu · Cumulative GPA: 3.9/4.0

B.Eng. in Information Security 2021-2025
Wuhan University Wuhan, China
Rank: 1/182 · Outstanding Graduate

Research & Internship Experience

Tencent AI Lab Feb. 2025 - May. 2025
Mentor(s): Tianqing Fang
Project 1: WebCoT — Enhancing Web Agent Reasoning by Reconstructing CoT (EMNLP 2025 Findings)
Project 2: Explore to Evolve — Scaling Evolved Aggregation Logic for Deep Research Agents (ACL 2026 Main)

MIT Media Lab Oct. 2024 - Jan. 2025
Supervisor: Prof. Paul Liang, Dr. Yi R. (May) Fung
Project: VLM2-Bench — Evaluating How Well VLMs Implicitly Link Matching Visual Cues (ACL 2025 Main)

Tong Zhang's Group, UIUC Feb. 2024 - Dec. 2025
Supervisor: Prof. Tong Zhang
Project 1: Image Textualization — Automatic Framework for Rich & Detailed Image Descriptions (NeurIPS 2024)
Project 2: Personalized Visual Instruction Tuning — PVIT framework & PVIT-3M dataset (ICLR 2025)
Project 3: MLLM-Protector — Ensuring MLLM's Safety without Hurting Performance (EMNLP 2024)
Project 4: Bridge-Coder — Overcoming Language Gaps in Low-Resource Code (ACL 2025 Findings)

Publications

*(◇ marks my first- or co-first-authored papers; * denotes equal contribution.)*

◇ **PROGRESSLM: Towards Progress Reasoning in Vision-Language Models**
Jianshu Zhang*, Chengxuan Qian*, Haosen Sun, Haoran Lu, Dingcheng Wang, Letian Xue, Han Liu
*ACL 2026 Main (**Oral**)*

◇ **Progress Reward Modeling for Robotic Learning: A Comprehensive Survey**
Jianshu Zhang*, Keliang Wu*, Haoran Lu*, Anbang Liu, Ce Zhang, Weijie Yin, Chengxuan Qian, Xiyuan Yang, Zhenyu Pan, Guo Ye, Han Liu
Tech Report arXiv:2607.21655

◇ **Video2Skill: From Streaming Experience to Reusable Embodied Skills**
Jianshu Zhang*, Ce Zhang*, Xiyuan Yang, Chenwei Xu, Haoran Lu, Yijiang Li, Yaqi Xie, Katia P. Sycara, Han Liu
Under Review 2026

◇ **ProgressCompass: Embodied Progress Reward Models Are Lost Without the Right Context**
Jianshu Zhang*, Keliang Wu*, Chengxuan Qian, Xiyuan Yang, Haoran Lu, Anbang Liu, Han Liu
Under Review 2026

◇ **SpaceNum: From Numerical Tokens to Visual Geometry**
Jianshu Zhang, Ce Zhang, Yijiang Li, Huifeixin Chen, Haoran Lu, Letian Xue, Bingyang Wang, Han Liu
Under Review 2026

◇ **MotionArena: Do Vision-Language Models Understand Fine-grained Manipulation Motion?**
Yiming Liang*, **Jianshu Zhang***, Ce Zhang*, Chenwei Xu, Weijie Yin, Han Liu
Under Review 2026

◇ **EISWorld: Can Models Predict the Consequences of Real-World Projects?**
Jianshu Zhang, Huifeixin Chen, Shang Wu, Zhenyu Pan, Aihan Liu, Aerith Y. Netzer, Kelsey Rydland, Han Liu, James Jaehoon Lee
Under Review 2026

- **StreamScout: Learning When to Look Deeper for Streaming Video Understanding**
Ce Zhang, Jing Bi, Jinxi He, **Jianshu Zhang**, Jingyang Lin, Yunzhong Xiao, Minghao Fu, Yaqi Xie, Katia P. Sycara, Ming Zhou
Under Review 2026

- **AnnotateAnything: Automatic Annotation of 3D Assets for Robot Manipulation**
Haoran Lu, Mutian Shen, Shuyang Yu, Yu Xiao, Songling Liu, **Jianshu Zhang**, Shang Wu, Yue Chen, Guo Ye, Jiayi Wang, Zhaoran Wang, Han Liu
Under Review 2026

- **MagicSim: A Unified Infrastructure for Executable Embodied Interaction**
Haoran Lu, Songling Liu, Yue Chen, Guo Ye, Mutian Shen, Shuyang Yu, Yu Xiao, Jihai Zhao, Shang Wu, **Jianshu Zhang**, Xiangtian Gui, Chuye Hong, Yuran Wang, Maojiang Su, Jiayi Wang, Ruihai Wu, Zhaoran Wang, Han Liu
Tech Report arXiv:2606.17511

- **MindCube: Spatial Mental Modeling from Limited Views**
Qineng Wang, Baiqiao Yin, Pingyue Zhang, **Jianshu Zhang**, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, Han Liu, Ranjay Krishna, Saining Xie, Jiajun Wu, Li Fei-Fei, Manling Li
*ICLR 2026 (also SP4V@ICCV 2025 **Best Paper Award**)*

- **Phys4D: Fine-Grained Physics-Consistent 4D Modeling from Video Diffusion**
Haoran Lu, Shang Wu, **Jianshu Zhang**, Maojiang Su, Guo Ye, Chenwei Xu, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Zhaoran Wang, Han Liu
Under Review 2026

- **Explore to Evolve: Scaling Evolved Aggregation Logic via Proactive Online Exploration for Deep Research Agents**
Rui Wang, Ce Zhang, Jun-Yu Ma, **Jianshu Zhang**, Hongru Wang, Yi Chen, Boyang Xue, Tianqing Fang, Zhisong Zhang, Hongming Zhang, Haitao Mi, Dong Yu, Kam-Fai Wong
ACL 2026 Main

- **AdvEvo-MARL: Shaping Internalized Safety through Adversarial Co-Evolution in Multi-Agent Reinforcement Learning**
Zhenyu Pan, Yiting Zhang, Zhuo Liu, Yolo Yunlong Tang, Zeliang Zhang, Haozheng Luo, Yuwei Han, **Jianshu Zhang**, Dennis Wu, Hong-Yu Chen, Haoran Lu, Haoyang Fang, Manling Li, Chenliang Xu, Philip S. Yu, Han Liu
ICML 2026

- **VideoCritic: Diagnosing and Localizing Reasoning Errors in Video-Language Models**
  Chenwei Xu, **Jianshu Zhang**, Shang Wu, Lie Lu, Pranav Maneriker, Fan Du, Manling Li, Han Liu
  VidLLMs@CVPR 2026

◇ **VLM2-Bench: A Closer Look at How Well VLMs Implicitly Link Explicit Matching Visual Cues**
  Jianshu Zhang*, Dongyu Yao*, Renjie Pi, Paul Pu Liang, Yi R. Fung
  ACL 2025 Main

- **WebCoT: Enhancing Web Agent Reasoning by Reconstructing Chain-of-Thought in Reflection, Branching, and Rollback**
  Minda Hu, Tianqing Fang, **Jianshu Zhang**, Jun-Yu Ma, Zhisong Zhang, Jingyan Zhou, Hongming Zhang, Haitao Mi, Dong Yu, Irwin King
  EMNLP 2025 Findings

◇ **Personalized Visual Instruction Tuning**
  Renjie Pi*, **Jianshu Zhang***, Tianyang Han, Jipeng Zhang, Rui Pan, Tong Zhang
  ICLR 2025

◇ **CAN: Leveraging Clients As Navigators for Generative Replay in Federated Continual Learning**
  Xuankun Rong*, **Jianshu Zhang***, Kun He, Mang Ye
  ICML 2025

◇ **Bridge-Coder: Transferring Model Capabilities from High-Resource to Low-Resource Programming Language**
  Jipeng Zhang*, **Jianshu Zhang***, Yuanzhe Li*, Renjie Pi, Rui Pan, Runtao Liu, Ziqiang Zheng, Tong Zhang
  ACL 2025 Findings

- **MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models**
  Young-Jun Lee, Byung-Kwan Lee, **Jianshu Zhang**, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi
  ICCV 2025

- **FairReason: Balancing Reasoning and Social Bias in MLLMs**
  Zhenyu Pan, Yutong Zhang, **Jianshu Zhang**, Haoran Lu, Haozheng Luo, Yuwei Han, Philip S. Yu, Manling Li, Han Liu
  T2FM@ICCV 2025

◇ **Image Textualization: An Automatic Framework for Generating Rich and Detailed Image Descriptions**
  Renjie Pi*, **Jianshu Zhang***, Jipeng Zhang, Rui Pan, Zhekai Chen, Tong Zhang
  NeurIPS 2024

◇ **MLLM-Protector: Ensuring MLLM's Safety without Hurting Performance**
  Renjie Pi*, Tianyang Han*, **Jianshu Zhang***, Yueqi Xie, Rui Pan, Qing Lian, Hanze Dong, Jipeng Zhang, Tong Zhang
  EMNLP 2024 Main

◇ **FIRST: Teach a Reliable Large Language Model through Efficient Trustworthy Distillation**
  KaShun Shum*, Minrui Xu*, **Jianshu Zhang***, Zixin Chen, Shizhe Diao, Hanze Dong, Jipeng Zhang, Muhammad Omer Raza
  EMNLP 2024 Main

◇ **CORE: Mitigating Catastrophic Forgetting in Continual Learning through Cognitive Replay**
Jianshu Zhang*, Yankai Fu*, Ziheng Peng*, Dongyu Yao, Kun He
*CogSci 2024 (**Oral**)*

◇ **FuzzLLM: A Novel and Universal Fuzzing Framework for Proactively Discovering Jailbreak Vulnerabilities in Large Language Models**
Dongyu Yao*, **Jianshu Zhang***, Ian G. Harris, Marcel Carlsson
ICASSP 2024

Honors & Awards

- Oral ACL 2026 (Top 3.3%) 2026
- McCormick School of Engineering Fellowship, Northwestern University 2025
- National Scholarship (Top 0.2% nationally) 2024
- Lei Jun Computer Breakthrough Award (50K RMB), Wuhan University 2025
- Best Paper Award SP4V@ICCV 2025 2025
- Best Paper Honorable Mention LAW@NeurIPS 2025 2025

Academic Service

- **Conference Reviewer**: ACL (Outstanding Review 2026), ICML (Silver Reviewer 2026), ICLR, NeurIPS, EMNLP, CVPR, ICCV, TMLR